

Heat Sink Thermal Model - Summary Report

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What I did

Took the given heat-sink script and turned it into a function so I could sweep TDP (30-250W), air velocity (0.5-15 m/s) and TIM conductivity (1-12 W/mK) instead of just one operating point. Generated 9,000 data points from that, trained two ML models on it, and ran a correlation/sensitivity check to see what actually drives temperature vs resistance.

Assumptions

- Kept the physics exactly as given (Sieder-Tate/Dittus-Boelter correlations, same resistance network). Only refactored the code to be callable in a loop.
- $R_{jc} = 0.2$ °C/W, same as the base script (the reference PDF says 0.1, kept the script's value since that's the base model I was told to use).
- Geometry, TIM thickness and air properties held fixed - only the three specified parameters vary.
- Used a full grid sweep instead of random sampling so it's reproducible.

Key results

- Reproduces the original script exactly at the base point (150W, 1 m/s, 4 W/mK): $R_{total} = 0.369$ °C/W, $T_j = 80.4$ °C.
- Across the full sweep: R_{total} ranges 0.16-0.44 °C/W, T_j ranges 29-107°C.
- One thing worth flagging: with this fin spacing, Reynolds number never crosses 2300 even at 15 m/s, so the turbulent-flow part of the model never actually gets used in this dataset.

ML model comparison

Target	Model	MAE	RMSE	R ²
R_{total}	Linear Regression (poly)	0.0077	0.011	0.89
R_{total}	Random Forest	~0	~0	1.00
T_j	Linear Regression (poly)	1.34	2.03	0.99
T_j	Random Forest	0.12	0.33	0.9997

Random Forest wins on both targets, expected given the non-linear terms in the physics. The perfect R_{total} score looked suspicious at first, so I tested it by training only on $TDP < 200W$ and predicting $TDP \geq 200W$ (data it never saw). R_{total} still scored perfectly since it doesn't depend on TDP at all, but T_j 's score dropped to -0.54 - confirming the model interpolates well but can't extrapolate past its training range.

What drives what

R_{total} is mostly controlled by air velocity (~89% importance), with TIM conductivity playing a smaller role (~11%), and TDP has basically zero effect on it. T_j on the other hand is almost entirely explained by TDP (~94%), since $T_j = T_{ambient} + TDP \times R_{total}$ and TDP swings a lot more than R_{total} does across the sweep. Correlation numbers back this up (V_{air} vs R_{total} : $r = -0.80$, TDP vs T_j : $r = 0.97$).

Limitations

- No noise added to the data - it's a clean formula, so accuracy numbers here will look better than they would against real measured data.
- Turbulent flow regime never gets tested in this sweep.
- Only 3 parameters varied - geometry and R_{jc} stay fixed, so this only applies to this specific heat sink.

What I'd improve next

- Sweep a wider V_{air} range or smaller fin spacing to actually see the turbulent branch.
- Add geometry parameters to make the surrogate more general.
- Since R_{total} doesn't depend on TDP, a smaller 2-input model for R_{total} plus the closed-form equation for T_j would probably be more efficient than one 3-input black box.